A. Division, Remainders, and Congruence

Divisibility

For integers d and n,

$$d \mid n \iff \text{there is a } k \in \mathbb{Z} \text{ such that } n = dk.$$

Here $\mathbb{Z}$ is the set of integers, d is a **divisor** (or factor), n is a **multiple** of d, and k is an integer **witness**. To prove $d \nmid n$, show that no integer witness can exist.

Division Algorithm

For every $a \in \mathbb{Z}$ and positive integer b, there are unique integers q, r such that

$$a = bq + r, \qquad 0 \le r < b.$$

The integer a is the **dividend**, b the **divisor**, q the **quotient**, and r the **remainder**. We write $q = a \operatorname{div} b$ and $r = a \bmod b$. The bound on r makes the answer canonical, even when a is negative.

Congruence

For $m \ge 2$,

$$x \equiv y \pmod{m} \iff m \mid (x - y).$$

The positive integer m is the **modulus**; we may also write $x \equiv_m y$. Equivalently, $x = y + mk$ for some $k \in \mathbb{Z}$, or x and y have the same least nonnegative remainder modulo m.

One relationship, three views

If $a = bq + r$ with $0 \le r < b$, then

$$a \bmod b = r \quad \text{and} \quad a \equiv r \pmod{b}.$$

The equation gives the quotient and remainder; congruence records only the remainder class.

Basic divisibility laws

For integers of the indicated form:

$$\begin{aligned} d \mid a,\ d \mid b &\implies d \mid (ma + nb), \quad m, n \in \mathbb{Z}, \\ d \mid a &\implies d \mid ac, \quad c \in \mathbb{Z}, \\ d \mid a,\ a \mid b &\implies d \mid b. \end{aligned}$$

Each follows by substituting integer witnesses and factoring out d.

Similar example: a positive dividend

Problem. For $a = 263$ and positive divisor $b = 17$, find the canonical integers q, r with $a = bq + r$ and $0 \le r < b$. State $a \operatorname{div} b$, $a \bmod b$, and the corresponding congruence. Justify the congruence from its definition.
Worked solution. Find all three views for 263 divided by 17. **Choose q:** $17(15) = 255$ is the largest multiple not exceeding 263. **Compute r:** $263 - 255 = 8$, so

$$263 = 17(15) + 8, \qquad 0 \le 8 < 17.$$

The bound $0 \le 8 < 17$ makes this canonical. Therefore

$$263 \operatorname{div} 17 = 15, \qquad 263 \bmod 17 = 8, \qquad 263 \equiv 8 \pmod{17}.$$

Congruence check: $263 - 8 = 255 = 17(15)$, so $17 \mid (263 - 8)$.
Answer. $q = 15$, $r = 8$; $263 \operatorname{div} 17 = 15$, $263 \bmod 17 = 8$, and $263 \equiv_{17} 8$.

Similar example: a negative dividend

Problem. For $a = -52$ and divisor $b = 9$, find the canonical quotient and remainder. Write $a = bq + r$ with $0 \le r < b$, and state the div, mod, and congruence forms.
Worked solution. Find the canonical quotient and remainder for -52 divided by 9. **Reject the first try:** $-52 = 9(-5) - 7$ has a negative remainder. **Adjust:** decrease the quotient by 1 and add 9 to the remainder:

$$-52 = 9(-6) + 2, \qquad 0 \le 2 < 9.$$

Now $0 \le 2 < 9$. Thus

$$-52 \operatorname{div} 9 = -6, \qquad -52 \bmod 9 = 2, \qquad -52 \equiv 2 \pmod{9}.$$

Answer. $q = -6$, $r = 2$; $-52 \operatorname{div} 9 = -6$, $-52 \bmod 9 = 2$, and $-52 \equiv_9 2$.

Common mistake

Do not truncate a negative quotient toward zero. Choose the unique q that makes $0 \le r < b$. A negative number may be congruent to a negative residue, but $a \bmod b$ is the least nonnegative one.

Can you do this from the definitions?

Given a and $b > 0$, find q, r; verify the remainder bound; state $a \operatorname{div} b$, $a \bmod b$, and a congruence; justify the congruence by showing that b divides a difference.

A.4: recover a number

Problem. An integer a satisfies $a \operatorname{div} 13 = -4$ and $a \bmod 13 = 9$. Find a and verify the required remainder bound.
Worked solution. Insert $q = -4, r = 9$ into the division equation $a = 13q + r$: $a = 13(-4) + 9 = -43$. The given remainder is legal because $0 \le 9 < 13$.
Answer. $a = -43$.

A.5: test a congruence

Problem. Decide whether $347 \equiv -13 \pmod{24}$. Justify directly from the definition, then give the least nonnegative residue of 347 modulo 24.
Worked solution. Subtract the two integers: $347 - (-13) = 360 = 24(15)$. Thus 24 divides the difference, proving the congruence. For the least nonnegative residue, divide: $347 = 24(14) + 11$, with $0 \le 11 < 24$.
Answer. The congruence is true, and $347 \bmod 24 = 11$.

B. Base-b Representations

Base-b expansion

For $b \ge 2$, the numeral $(a_k a_{k-1} \cdots a_0)_b$ represents

$$a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0, \qquad 0 \le a_i < b.$$

Binary uses digits $0, 1$; octal uses 0 through 7; hexadecimal uses 0 through 9 and $A = 10, \ldots, F = 15$.

Place value, shifting, and digits

$$(10)_b = b, \qquad (100)_b = b^2, \qquad (10)_b (10)_b = (100)_b.$$

More generally,

$$\big((10)_b\big)^k = \big(1\underbrace{00\cdots0}_{k \text{ zeros}}\big)_b = b^k.$$

If $N = (a_k \cdots a_0)_b$ and $0 \le r < b$, then

$$bN = (a_k \cdots a_0 0)_b, \qquad bN + r = (a_k \cdots a_0 r)_b.$$

Equivalently,

$$(a_k \cdots a_1 a_0)_b = b(a_k \cdots a_1)_b + a_0.$$

Thus multiplying by b shifts left, while dividing by b removes the final digit: $\operatorname{div} b$ gives the remaining prefix and $\bmod b$ gives the removed digit.

Decimal to another base

Repeatedly divide by the new base. Record each remainder. The digits are the remainders read from *bottom to top*. Verify by expanding in powers of the base. Each line $n = bq + r$ reverses one shift-and-append step. In hexadecimal, remainders 10–15 become A–F.
Problem (B.4). Convert 1000 from decimal to hexadecimal using repeated division. Check by expanding in powers of 16.
Worked solution. The divisions are $1000 = 16(62) + 8$, $62 = 16(3) + 14$, and $3 = 16(0) + 3$. Read the remainders upward, replacing 14 with E. Check: $3(16^2) + 14(16) + 8 = 1000$.
Answer. $1000_{10} = (3E8)_{16}$.

Similar example: 202_{10} to binary

Problem. Convert the decimal integer 202 to binary by repeated division by 2. Show the quotients and remainders, explain the digit order, and verify the result using powers of 2.
Worked solution. Divide and record remainders:

$$\begin{aligned} 202 &= 2(101) + 0 & 101 &= 2(50) + 1 \\ 50 &= 2(25) + 0 & 25 &= 2(12) + 1 \\ 12 &= 2(6) + 0 & 6 &= 2(3) + 0 \\ 3 &= 2(1) + 1 & 1 &= 2(0) + 1 \end{aligned}$$

Read upward: the remainders in reverse chronological order give $202_{10} = 11001010_2$. Rebuilding by shifting and appending verifies the digit order:

$$\begin{aligned} &(1)_2 \to (11)_2 \to (110)_2 \\ &\quad \to (1100)_2 \to (11001)_2 \to (110010)_2 \\ &\quad \to (1100101)_2 \to (11001010)_2. \end{aligned}$$

Place-value check: $128 + 64 + 8 + 2 = 202$.
Answer. $202_{10} = (11001010)_2$.

Similar example: evaluate a numeral

Problem. Convert $(2301)_5$ to decimal using place values. Verify that every digit is legal in base 5.
Worked solution. Multiply each digit by the matching power of 5:

$$(2301)_5 = 2(5^3) + 3(5^2) + 0(5) + 1 = 326.$$

Thus the value is 326_{10}. The subscript names the base; it is not a multiplier.
Answer. All digits are in $\{0, 1, 2, 3, 4\}$, and $(2301)_5 = 326_{10}$.

Binary as a bridge

One hexadecimal digit equals exactly four binary bits; one octal digit equals exactly three. Pad with leading zeros, then regroup from the *right*.
Problem (B.3). Convert $(725)_8$ to binary and decimal. Keep three binary digits for each octal digit.
Worked solution. Convert $7, 2, 5$ separately to $111, 010, 101$. Check the value using octal place values: $7(8^2) + 2(8) + 5 = 469$.
Answer. $(725)_8 = (111\ 010\ 101)_2 = 469_{10}$.
Problem (B.5). Convert $(101101011)_2$ to octal and hexadecimal without first converting the whole number to decimal. Explain the grouping rule.
Worked solution. From the right, group into threes for octal: $(101\ 101\ 011)_2$. Group into fours for hexadecimal, adding leading zeros: $(0001\ 0110\ 1011)_2$. Each group is one digit because $8 = 2^3$ and $16 = 2^4$.
Answer. $(101101011)_2 = (553)_8 = (16B)_{16}$.

Similar example: hexadecimal to binary and octal

Problem. Convert $(2D7)_{16}$ to binary and octal, where $D = 13$. Use four bits per hexadecimal digit, regroup from the right for octal, and check the value in decimal.
Worked solution. Convert $(2D7)_{16}$. **Hex to binary:** replace each digit with four bits:

$$2 = 0010_2, \qquad D = 1101_2, \qquad 7 = 0111_2,$$

so

$$(2D7)_{16} = (0010\ 1101\ 0111)_2.$$

Binary to octal: regroup from the right in threes,

$$001\ 011\ 010\ 111 \implies (1327)_8.$$

Decimal check:

$$2(16^2) + 13(16) + 7 = 727.$$

Leading zeros change grouping, not value.
Answer. $(2D7)_{16} = (0010\,1101\,0111)_2 = (1327)_8 = 727_{10}$.

Similar example: the last two base-b digits

Problem. Let $n = (25314)_7$. Find $n \bmod 7^2$ without evaluating the entire numeral in decimal. Explain how the last two base-seven digits determine the answer, and report the residue in decimal.
Worked solution. Let $n = (25314)_7$. Then

$$n = 2(7^4) + 5(7^3) + 3(7^2) + 1(7) + 4.$$

Discard higher places: every term containing 7^2 is congruent to 0 modulo 7^2. Therefore

$$n \bmod 7^2 = 1(7) + 4 = 11.$$

Interpret: the last two base-seven digits are $(14)_7$, whose decimal value is $1(7) + 4 = 11$, not 14.
Answer. $n \bmod 49 = 11 = (14)_7$.

Conversion checks

Digits must be legal for the base. Binary/hex groups have four bits; binary/octal groups have three. Expand the final numeral to verify it represents the same integer.

C–D. Modular Arithmetic, GCD, and Bézout

Arithmetic with congruences

If $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then

$$a + c \equiv b + d \pmod m, \qquad ac \equiv bd \pmod m.$$

In particular, $a^n \equiv b^n \pmod m$ for every positive integer n. The exact remainder identities are

$$(a + c) \bmod m = ((a \bmod m) + (c \bmod m)) \bmod m,$$
$$ac \bmod m = ((a \bmod m)(c \bmod m)) \bmod m.$$

Reduce early; report a least nonnegative residue for $x \bmod m$.

Similar example: reduce early

Problem. Compute $\big((-47) \cdot 38 + 19\big) \bmod 12$. Reduce the inputs before combining them and give the least nonnegative residue.
Worked solution. Compute the least nonnegative residue of $(-47) \cdot 38 + 19$ modulo 12. **Reduce each input:**

$$-47 \equiv 1, \quad 38 \equiv 2, \quad 19 \equiv 7 \pmod{12},$$

Operate on the small representatives:

$$(-47) \cdot 38 + 19 \equiv 1(2) + 7 \equiv 9 \pmod{12}.$$

Since $0 \le 9 < 12$, $((-47) \cdot 38 + 19) \bmod 12 = 9$.
Answer. $\big((-47) \cdot 38 + 19\big) \bmod 12 = 9$.

Powers modulo m: cycles and squaring

Problem (C.1(b)). Compute $3^{47} \bmod 10$, giving the least nonnegative residue and showing a cycle that avoids large arithmetic.
Worked solution: cycle method. Compute residues until they repeat. Since $3, 9, 7, 1$ repeat for powers of 3 modulo 10, and $47 \equiv 3 \pmod 4$,

$$3^{47} \equiv 3^3 \equiv 7 \pmod{10}.$$

Answer. $3^{47} \bmod 10 = 7$.
Problem (C.4). Compute $5^{117} \bmod 13$ by repeated squaring. Display the powers used in the final product.
Worked solution: repeated squaring. Square and reduce at every step. Modulo 13,

$$5^2 \equiv -1, \qquad 5^4 \equiv 1,$$

so all powers $5^{4j} \equiv 1$. Because $117 = 64 + 32 + 16 + 4 + 1$,

$$5^{117} \equiv 1 \cdot 1 \cdot 1 \cdot 1 \cdot 5 \equiv 5 \pmod{13}.$$

Answer. $5^{117} \bmod 13 = 5$. State the cycle length or the exponent decomposition you used.

GCD and relative primality

For integers a, b, not both zero, $\gcd(a, b)$ is the greatest positive integer dividing both. The numbers are **relatively prime** when $\gcd(a, b) = 1$. The numbers themselves need not be prime.

Euclidean algorithm

Put the larger positive input first, then repeatedly use

$$a = bq + r, \qquad 0 \le r < b,$$

then replace (a, b) by (b, r). Stop when the remainder is 0. The last *nonzero* remainder is the gcd. Indeed, from $a = bq + r$, the common divisors of a, b are exactly those of $b, r = a - bq$, so $\gcd(a, b) = \gcd(b, r)$. Swap the inputs first if necessary.

Similar example: Euclidean algorithm and Bézout

Problem. Use the Euclidean algorithm to find $\gcd(246, 102)$. Then back-substitute to find integers s, t such that $246s + 102t = \gcd(246, 102)$. Verify the final identity.
Worked solution. Find $\gcd(246, 102)$ and express it as $246s + 102t$. **Step 1: run the Euclidean algorithm.**

$$\begin{aligned} 246 &= 2(102) + 42, \\ 102 &= 2(42) + 18, \\ 42 &= 2(18) + 6, \\ 18 &= 3(6) + 0. \end{aligned}$$

The last nonzero remainder is 6, so it is the gcd. **Step 2: back-substitute.** Bézout's identity says that $\gcd(a, b) = sa + tb$ for some $s, t \in \mathbb{Z}$. Back-substitute:

$$\begin{aligned} 6 &= 42 - 2(18) \\ &= 42 - 2\big(102 - 2(42)\big) \\ &= 5(42) - 2(102) \\ &= 5\big(246 - 2(102)\big) - 2(102) \\ &= 5(246) - 12(102). \end{aligned}$$

Thus $s = 5$ and $t = -12$. **Step 3: verify both conclusions:**

$$6 \mid 246, \quad 6 \mid 102, \qquad 5(246) - 12(102) = 6.$$

Answer. $\gcd(246, 102) = 6$, with $s = 5$ and $t = -12$.

Bézout and relative-prime checks

Back-substitute through every displayed Euclidean equation; preserve the original input order in the final identity. Coefficients are not unique, so verify them by substitution.
Similar example — Problem. Are the positive integers 16 and 45 relatively prime? Justify your answer using their prime factorizations.
Worked solution. We have $16 = 2^4$ and $45 = 3^2 \cdot 5$. Their prime factorizations share no prime factor, so their only positive common divisor is 1.
Answer. $\gcd(16, 45) = 1$, so 16 and 45 are relatively prime. Composite numbers can be relatively prime.

E. Definition-Driven Proofs and Quiz Bridges

Even, odd, prime, and composite

An integer is **even** if it is $2k$, and **odd** if it is $2k + 1$, for some $k \in \mathbb{Z}$. A positive integer $p > 1$ is **prime** if its only positive divisors are 1 and p; an integer greater than 1 that is not prime is **composite**.

E.1: divisibility and linear combinations

Problem. Let $d, a, b, m, n \in \mathbb{Z}$. Suppose $d \mid a$ and $d \mid b$. Prove directly from the definition that $d \mid (ma + nb)$. State where each hypothesis is used.
Worked solution. Unpack: assume $d \mid a$ and $d \mid b$, so $a = dr$ and $b = ds$ for some $r, s \in \mathbb{Z}$. **Substitute and factor:** for any $m, n \in \mathbb{Z}$,

$$ma + nb = m(dr) + n(ds) = d(mr + ns).$$

Close: $mr + ns \in \mathbb{Z}$, so the display gives the required witness and $d \mid (ma + nb)$.
Answer. The two hypotheses give $a = dr, b = ds$; the integer witness $mr + ns$ proves $d \mid (ma + nb)$.

Similar example: a modulus that factors

Problem. Let $p, q \in \mathbb{Z}$ with $p, q \geq 2$, let $d = pq$, and let $x, y \in \mathbb{Z}$. Suppose $x \equiv_d y$. Prove directly that $x \equiv_p y$ and $x \equiv_q y$. Are p, q required to be prime?
Worked solution. Unpack: suppose $p, q \geq 2$, $d = pq$, and $x \equiv y \pmod{d}$. Then $d \mid (x - y)$, so $x - y = kd = kpq$ for some $k \in \mathbb{Z}$. **Refactor:**

$$x - y = p(kq) \quad \text{and} \quad x - y = q(kp).$$

Close: $kq, kp \in \mathbb{Z}$, so $p \mid (x - y)$ and $q \mid (x - y)$, giving

$$x \equiv y \pmod{p}, \qquad x \equiv y \pmod{q}.$$

This direction needs only positive moduli with $p \mid d$ and $q \mid d$; primality is not required. For instance, $103 - 4 = 99$ proves the same claim from modulus 33 to moduli 3 and 11.
Answer. Both congruences follow from witnesses kq and kp. Primality is not required.

A reliable proof skeleton

1. Translate each hypothesis using its definition.
2. Name the integer witnesses.
3. Manipulate the resulting equations.
4. Exhibit the witness required by the conclusion.
5. State why the witness is an integer.

C.5: products preserve congruence

Problem. Suppose $a \equiv b \pmod m$ and $c \equiv d \pmod m$, where $a, b, c, d \in \mathbb{Z}$ and $m \geq 2$ is an integer. Prove from the definition that $ac \equiv bd \pmod m$.
Worked solution. Unpack: assume $a \equiv b \pmod m$ and $c \equiv d \pmod m$. Then $a = b + mk$ and $c = d + m\ell$ for some $k, \ell \in \mathbb{Z}$. **Expand and factor:**

$$ac - bd = m(b\ell + dk + mk\ell).$$

Close: the parenthesized expression is an integer, so $m \mid (ac - bd)$ and $ac \equiv bd \pmod m$.
Answer. $ac - bd = m(b\ell + dk + mk\ell)$ exhibits an integer multiple of m, so $ac \equiv_m bd$.

E.2: odd squares modulo 8

Problem. Let $n \in \mathbb{Z}$ be odd. Prove $n^2 \equiv 1 \pmod 8$. Begin with the definition of odd and end by exhibiting an integer multiplier of 8.
Worked solution. Represent the odd integer: let $n = 2k + 1$. **Expand:**

$$n^2 - 1 = 4k(k + 1).$$

Find the extra factor of 2: one of $k, k + 1$ is even, so $k(k + 1) = 2t$ for some $t \in \mathbb{Z}$. Hence $n^2 - 1 = 8t$. Thus $8 \mid (n^2 - 1)$, proving $n^2 \equiv 1 \pmod 8$.
Answer. $n^2 - 1 = 8t$ for an integer t, so $n^2 \equiv_8 1$.

E.3: transitivity

Problem. Let $a, b, c \in \mathbb{Z}$. Prove directly that if $a \mid b$ and $b \mid c$, then $a \mid c$. Use distinct integer witnesses for the two hypotheses.
Worked solution. By $a \mid b$, write $b = ar$ for some $r \in \mathbb{Z}$. By $b \mid c$, write $c = bs$ for some $s \in \mathbb{Z}$. Substitute: $c = (ar)s = a(rs)$. Since $rs \in \mathbb{Z}$, this equation supplies the witness required by $a \mid c$.
Answer. $a \mid c$, with integer witness rs.

E.5: three consecutive integers

Problem. Prove that the product of any three consecutive integers is divisible by 6. Exhibit factors of 2 and 3, then combine their integer witnesses to produce an integer factor of 6.
Worked solution. Let $n \in \mathbb{Z}$ be arbitrary and set $P = n(n + 1)(n + 2)$. Of two consecutive integers one is even; thus at least one factor of P is even and $P = 2r$ for some $r \in \mathbb{Z}$. The residues of the three consecutive factors modulo 3 are $0, 1, 2$ in some order, so $P = 3s$ for some $s \in \mathbb{Z}$. Combine the two witnesses:

$$P = 3P - 2P = 3(2r) - 2(3s) = 6(r - s).$$

Since $r - s \in \mathbb{Z}$, the definition proves $6 \mid P$.
Answer. $6 \mid n(n + 1)(n + 2)$ for every integer n.

Claims that need an extra hypothesis

From $a \mid bc$, it does *not* follow in general that $a \mid b$ or $a \mid c$: $6 \mid 2 \cdot 3$, but 6 divides neither factor. The implication is valid when a is prime (Euclid's lemma): if prime $p \mid bc$, then $p \mid b$ or $p \mid c$.

Core readiness and format

A: canonical division and congruence. **B:** base conversion. **C:** modular arithmetic, cycles, squaring, and congruence proofs. **D:** Euclidean algorithm and Bézout. **E:** direct proofs and counterexamples. A live Core designates one problem from each section A–E and lasts 20 minutes. Four fully correct problems earns **Meets**; all five earns **Exceeds**. Every retake is a complete new form. Show enough work for each answer and proof to be checked; approved accommodations apply.